

Kiran Chhatre

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Research Summary

Research scientist working on multimodal world models, video generation and understanding, embodied AI, and 3D human behavior. My experience spans reinforcement learning and preference optimization for vision language models, controllable video generation, physics-based synthetic data, open vocabulary 3D vision, human motion, and robot perception.

Experience

Meta

Research Scientist Intern — World Models

Sausalito, California

Apr 2026–Sep 2026

- Investigate world models for representing people, interactions, future behavior, and evolving social states.
- Study RL and DPO for improving temporal understanding and reasoning in foundation vision language models.

Adobe Research

Research Scientist Intern — Video Generation & 3D Vision

London, UK / Bengaluru, India

Apr 2024–Mar 2026

- Led TrajectoryMover, a controllable video-to-video framework for scene-aware object relocation from sparse spatial controls.
- Built physics-based paired video data using Blender/Bullet Physics, with scene modifications and collision-aware trajectories.
- Developed 3D texture-aware diffusion representations for open-vocabulary clothing and body-part parsing.

Sony AI

Ph.D. Intern — Video Understanding & Creative AI

Kyoto, Japan / Remote

Apr 2025–Mar 2026

- Developed a multimodal video understanding framework for temporal localization and dense action captioning of human motion, with LLM-as-a-judge evaluation.

Electronic Arts SEED

Ph.D. Intern — Interactive Agents & Digital Humans

Stockholm, Sweden

Oct 2024–Mar 2025

- Researched generation and evaluation of expressive digital humans and interactive behavior in 3D environments.

Max Planck Institute for Intelligent Systems

Guest Scientist — Generative Human Motion

Tübingen, Germany

Sep 2022–Dec 2024

- Developed a latent diffusion model that disentangles speech content, emotion, and style for controllable 3D human motion.

Lawrence Berkeley National Laboratory

Research Affiliate — Multi-Agent Simulation

Berkeley, California

Nov 2019–Oct 2020

- Developed distributed Bayesian optimization methods for large-scale multi-agent urban transportation simulations.

McKinsey & Co. Learning Center

Robotics Developer

Aachen, Germany

Apr 2018–Mar 2019

- Developed SLAM, robot-vision, and automated welding and assembly workflows for industrial ABB and KUKA robots.

RoboCup Logistics League

Robotics Developer — Carologistics Team

Aachen, Germany

Nov 2018–Mar 2019

- Developed markerless machine recognition for autonomous production tasks in the RoboCup Logistics League.

Education

KTH Royal Institute of Technology, Stockholm, Sweden

Ph.D. in Computer Science, 2021–2026

RWTH Aachen University, Aachen, Germany

M.Sc. in Mechanical Engineering, 2017–2020

COEP Technological University, Pune, India

B.Tech. in Mechanical Engineering, 2010–2014

Selected Publications

K. Chhatre et al.. "TrajectoryMover: Generative Movement of Object Trajectories in Videos." *arXiv*, 2026

K. Chhatre et al.. "Learning 3D Texture-Aware Representations for Human Clothing and Body-Part Parsing." *AAAI*, 2026

K. Chhatre et al.. "Emotional Speech-Driven 3D Body Animation via Disentangled Latent Diffusion." *CVPR*, 2024

K. Chhatre et al.. "Audiopedia: Audio QA with Knowledge." *ICASSP Oral*, 2025

H. Du et al.. "Evaluating Gesture and Voice for Emotion and Empathy in VR and 2D." *ACM IVA Best Paper*, 2025

R. Daněček et al.. "Emotional Speech-Driven Animation with Content-Emotion Disentanglement." *SIGGRAPH Asia*, 2023

Technical Skills

Models: world models, vision-language models, video diffusion, latent diffusion, autoregressive models, RL, DPO

Video & Multimodal: video temporal localization, controllable generation, multimodal evaluation, LLM-as-a-judge

Embodied AI & 3D: robot perception, SLAM, synthetic data, physics simulation, multi-agent systems, 3D human motion

Engineering: Python, PyTorch, Ray, Blender/bpy, Bullet Physics, PyTorch3D, Open3D, Unreal Engine, FFmpeg, AWS, Git